BLOCKCHAIN ENGINEERING PRACTICAL-1

ROLL NO: 2203A51377

BATCH NO:30

Objective:

To learn blockchain interaction by creating a cryptocurrency wallet, checking wallet balance, and simulating transactions using Python and Web3

Requirements:

- Install Python 3.x
- Set up VS Code with Python extension
- Install required Python libraries:
- pip install web3
- Use a test blockchain network (Ethereum Sepolia / Ganache local blockchain)
- Basic understanding of blockchain wallets and private keys

Practical Description:

Step 1: Environment Setup

- Install Python and VS Code
- Install Web3.py library
- Create a Python file named wallet_interaction.py

Step 2: Wallet and Blockchain Interaction Script

Create a Python script that:

- Connects to a blockchain network
- Loads a wallet using a private key
- Fetches wallet address
- Checks wallet balance
- Demonstrates transaction preparation (without real funds)

CODE:

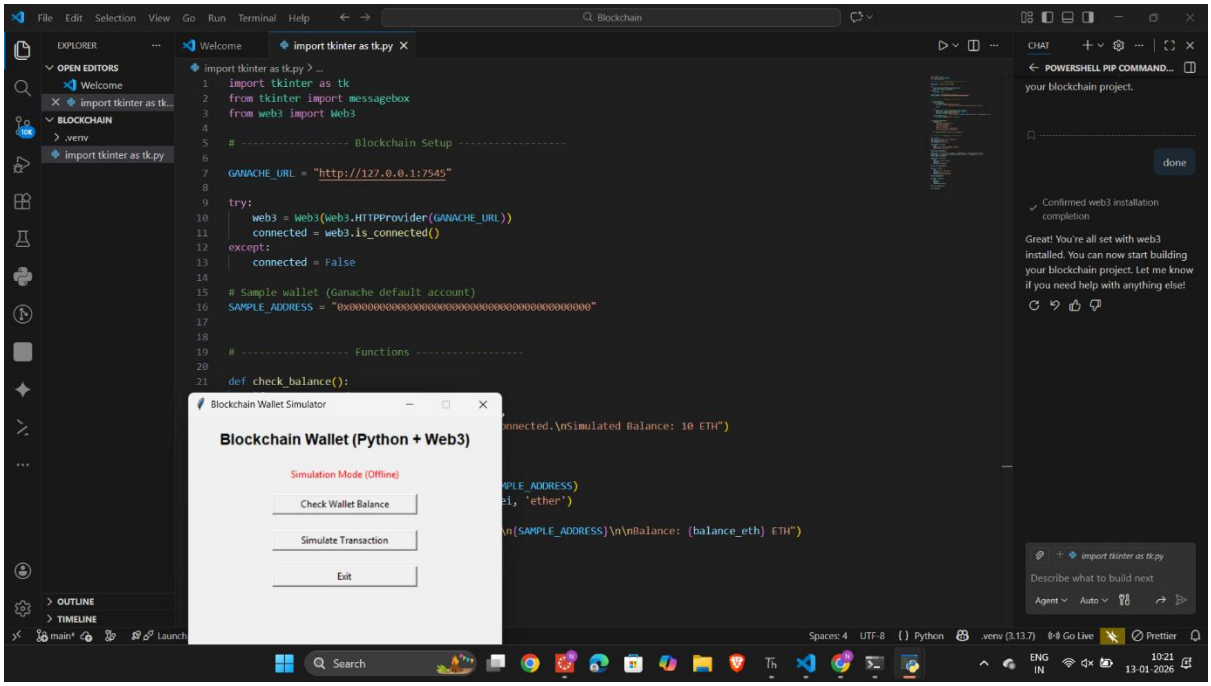
```
import tkinter as tk

from tkinter import messagebox

from web3 import Web3

# ---------------- Blockchain Setup ----------------

GANACHE_URL = "http://127.0.0.1:7545"

try:

    web3 = Web3(Web3.HTTPProvider(GANACHE_URL))

    connected = web3.is_connected()

except:

    connected = False

# Replace with your Ganache account address if connected

SAMPLE_ADDRESS = "0x0000000000000000000000000000000000000000"

# ---------------- Functions ----------------

def check_balance():

    if connected:

        try:

            wei_balance = web3.eth.get_balance(SAMPLE_ADDRESS)

            balance_eth = web3.from_wei(wei_balance, 'ether')

            formatted_balance = f"{balance_eth:.2f}"


            messagebox.showinfo(

                "Wallet Balance",

                f"Connected to Ganache\n\n"

                f"Wallet Address:\n{SAMPLE_ADDRESS}\n\n"

                f"Balance: {formatted_balance} ETH"

            )
```

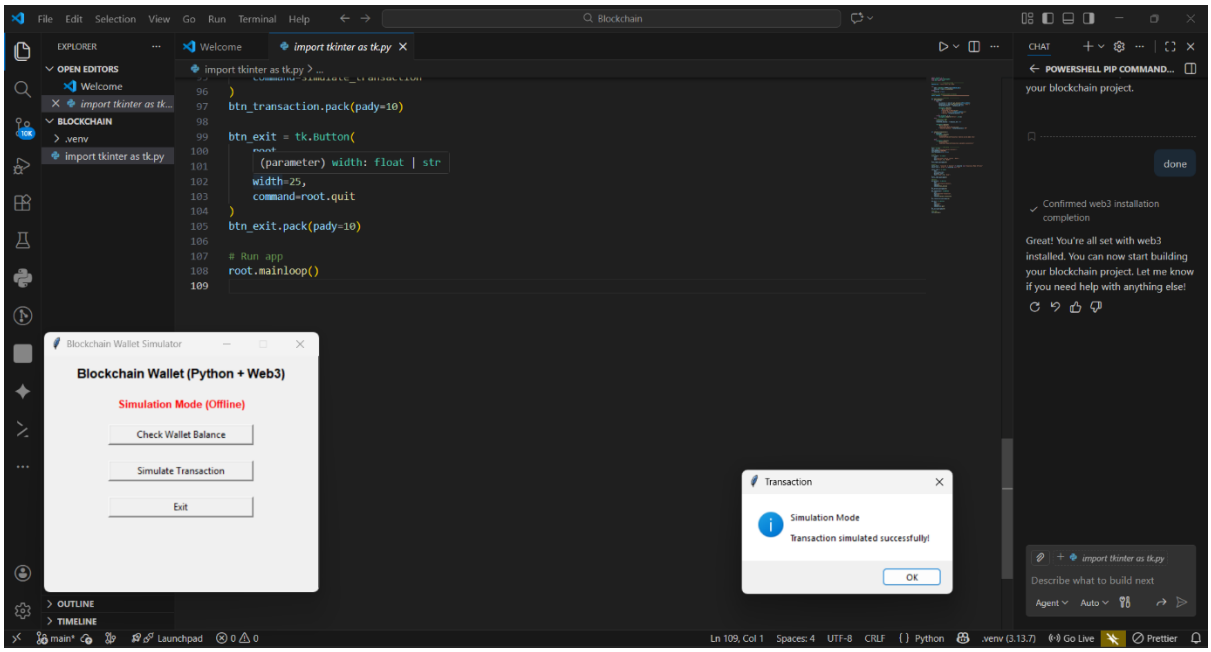
```
        except Exception as e:
            messagebox.showerror("Error", str(e))
    else:
        # Simulation Mode
        balance_eth = 10
        formatted_balance = f"{balance_eth:.2f}"

        messagebox.showinfo(
            "Wallet Balance",
            f"Simulation Mode (Offline)\n\n"
            f"Simulated Balance: {formatted_balance} ETH"
        )

def simulate_transaction():
    if connected:
        messagebox.showinfo(
            "Transaction",
            "Connected Mode\n\nTransaction feature can be added here."
        )
    else:
        messagebox.showinfo(
            "Transaction",
            "Simulation Mode\n\nTransaction simulated successfully!"
        )

# ---------------- GUI Setup ----------------
root = tk.Tk()
root.title("Blockchain Wallet Simulator")
```

```
root.geometry("350x300")
root.resizable(False, False)

# Title
title_label = tk.Label(
    root,
    text="Blockchain Wallet (Python + Web3)",
    font=("Arial", 12, "bold")
)
title_label.pack(pady=10)

# Status
status_text = "Connected to Ganache" if connected else "Simulation Mode (Offline)"
status_color = "green" if connected else "red"

status_label = tk.Label(
    root,
    text=status_text,
    fg=status_color,
    font=("Arial", 10, "bold")
)
status_label.pack(pady=5)

# Buttons
btn_balance = tk.Button(
    root,
    text="Check Wallet Balance",
    width=25,
```

```
    command=check_balance
)
btn_balance.pack(pady=10)

btn_transaction = tk.Button(
    root,
    text="Simulate Transaction",
    width=25,
    command=simulate_transaction
)
btn_transaction.pack(pady=10)

btn_exit = tk.Button(
    root,
    text="Exit",
    width=25,
    command=root.quit
)
btn_exit.pack(pady=10)

# Run app
root.mainloop()
```

OUTPUT:


import tkinter as tk
from tkinter import messagebox
from web3 import Web3
# ---------- Blockchain Setup ----------
GANACHE_URL = "http://127.0.0.1:7545"
try:
web3 = Web3(Web3.HTTPProvider(GANACHE_URL))
connected = web3.is_connected()
except:
connected = False
# Sample wallet (Ganache default account)
SAMPLE_ADDRESS = "0x0000000000000000000000000000000000000000"
# ---------- Functions ----------
def check_balance():
Blockchain Wallet Simulator
Blockchain Wallet (Python + Web3)
Simulation Mode (Offline)
Check Wallet Balance
Simulate Transaction
Exit
done
Great! You're all set with web3 installed. You can now start building your blockchain project. Let me know if you need help with anything else!


btn_transaction.pack(pady=10)
btn_exit = tk.Button(
(parameter) width: float | str
width=25,
command=root.quit
btn_exit.pack(pady=10)
# Run app
root.mainloop()
Blockchain Wallet Simulator
Blockchain Wallet (Python + Web3)
Simulation Mode (Offline)
Check Wallet Balance
Simulate Transaction
Exit
Transaction
Simulation Mode
Transaction simulated successfully!
OK
done
Great! You're all set with web3 installed. You can now start building your blockchain project. Let me know if you need help with anything else!